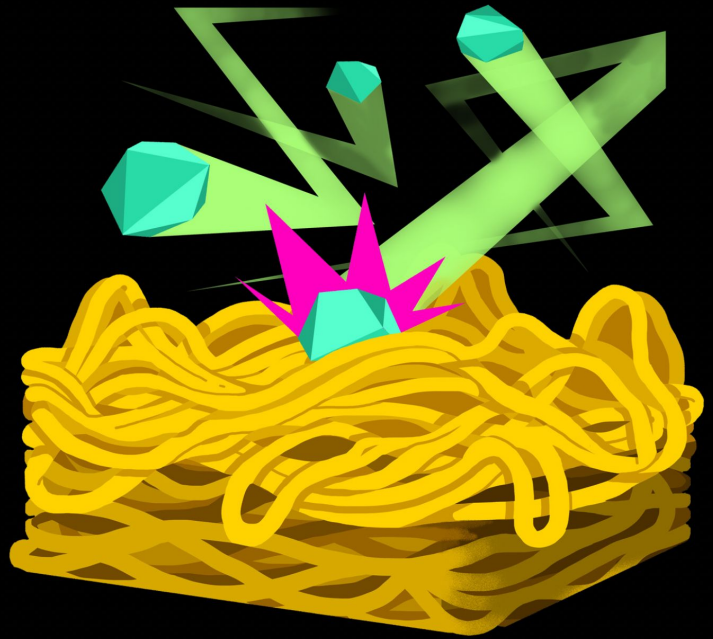


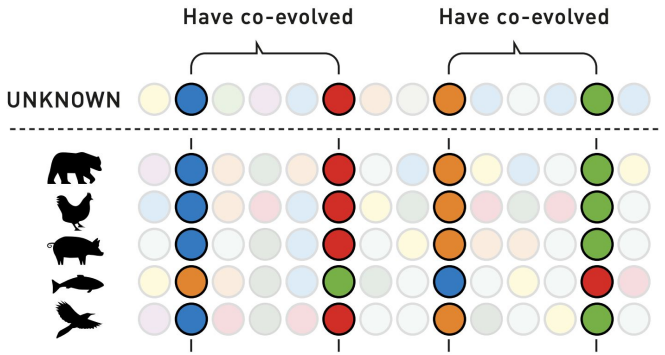
Exploring the Energy Landscape of the “Spaghetti Protein” via Optimal MSA Selection

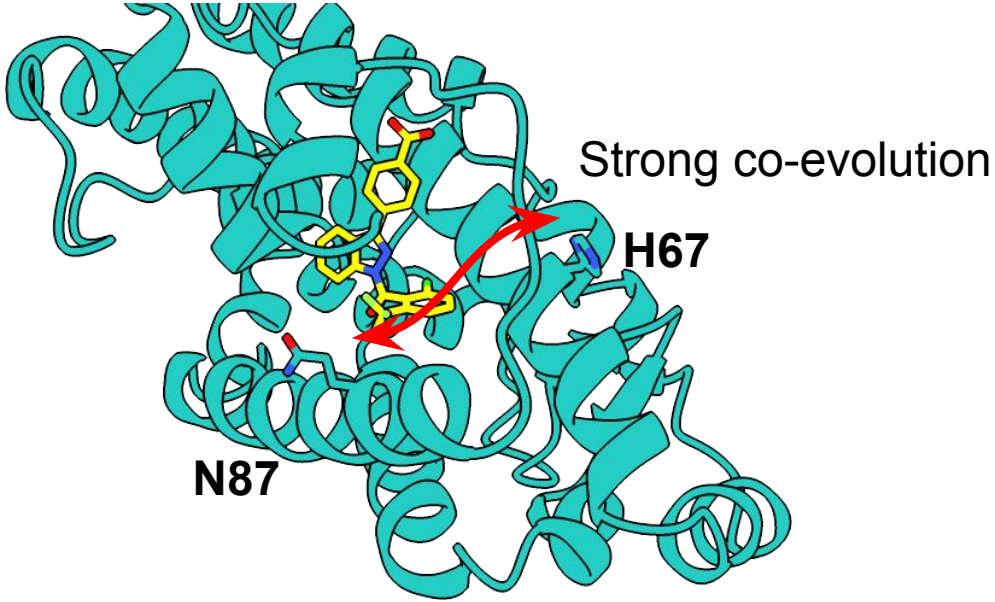
- Spaghetti Team -

Disclaimer:
Finetuning-Free Approach

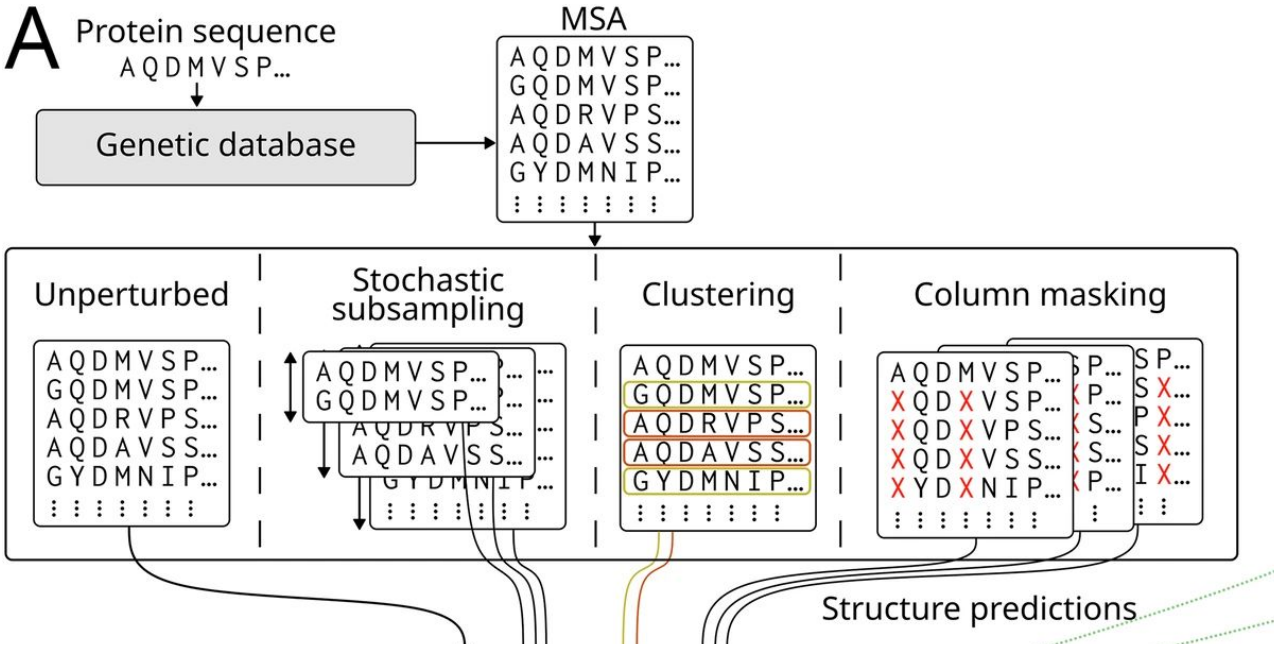
- Spaghetti Team -

Co-evolution in MSA





Weaken the co-evolution



OpenFold 3

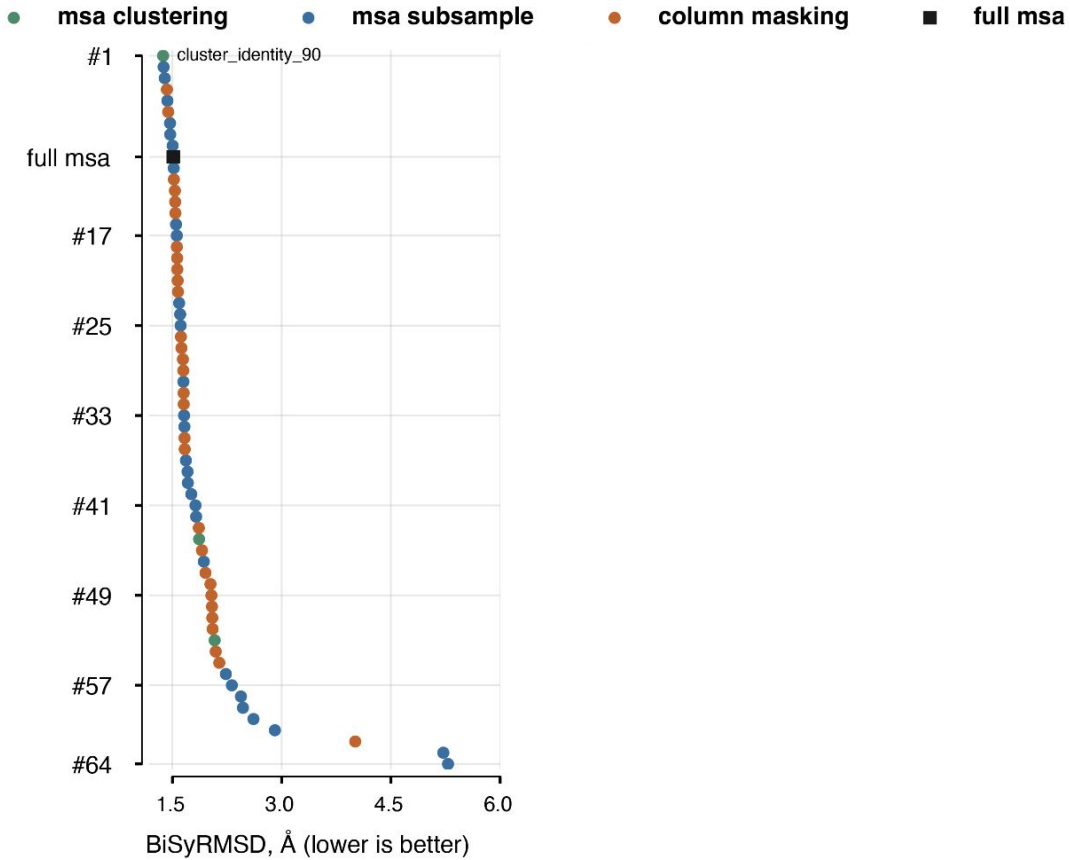
https://www.nature.com/articles/s42004-026-02198-x

Find a “perfect”MSA

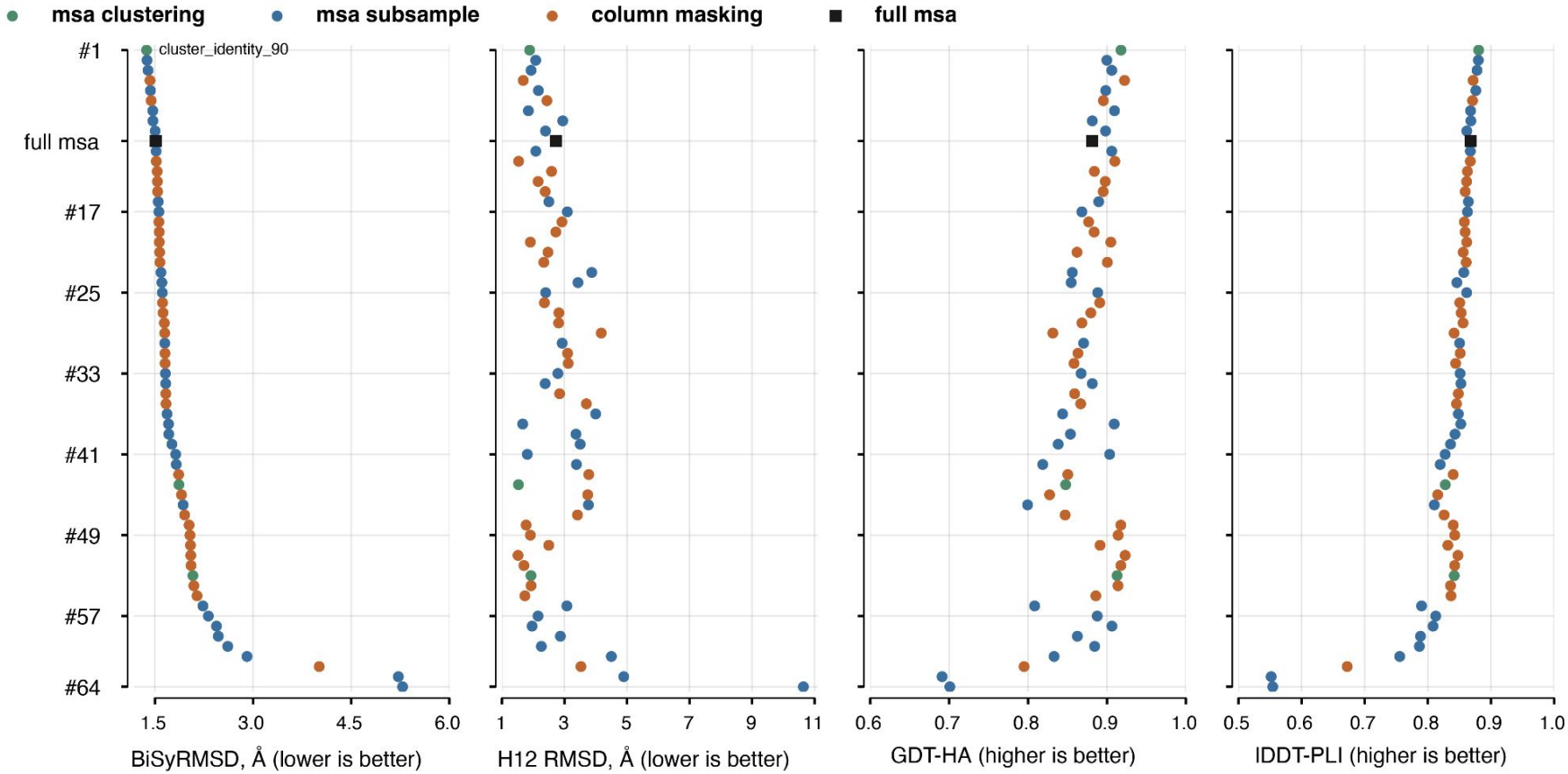
- Full MSA
- Subsampling randomly at 32, 64, 128 sequences x 10 times
- Column masking MSA at 5, 10, 15% x 10 times
- Clustering the MSA and 90, 70, 50 identity

Predict all for 77 ligands in the training datasets total: 4928 predictions

Did we find the “perfect” MSA?



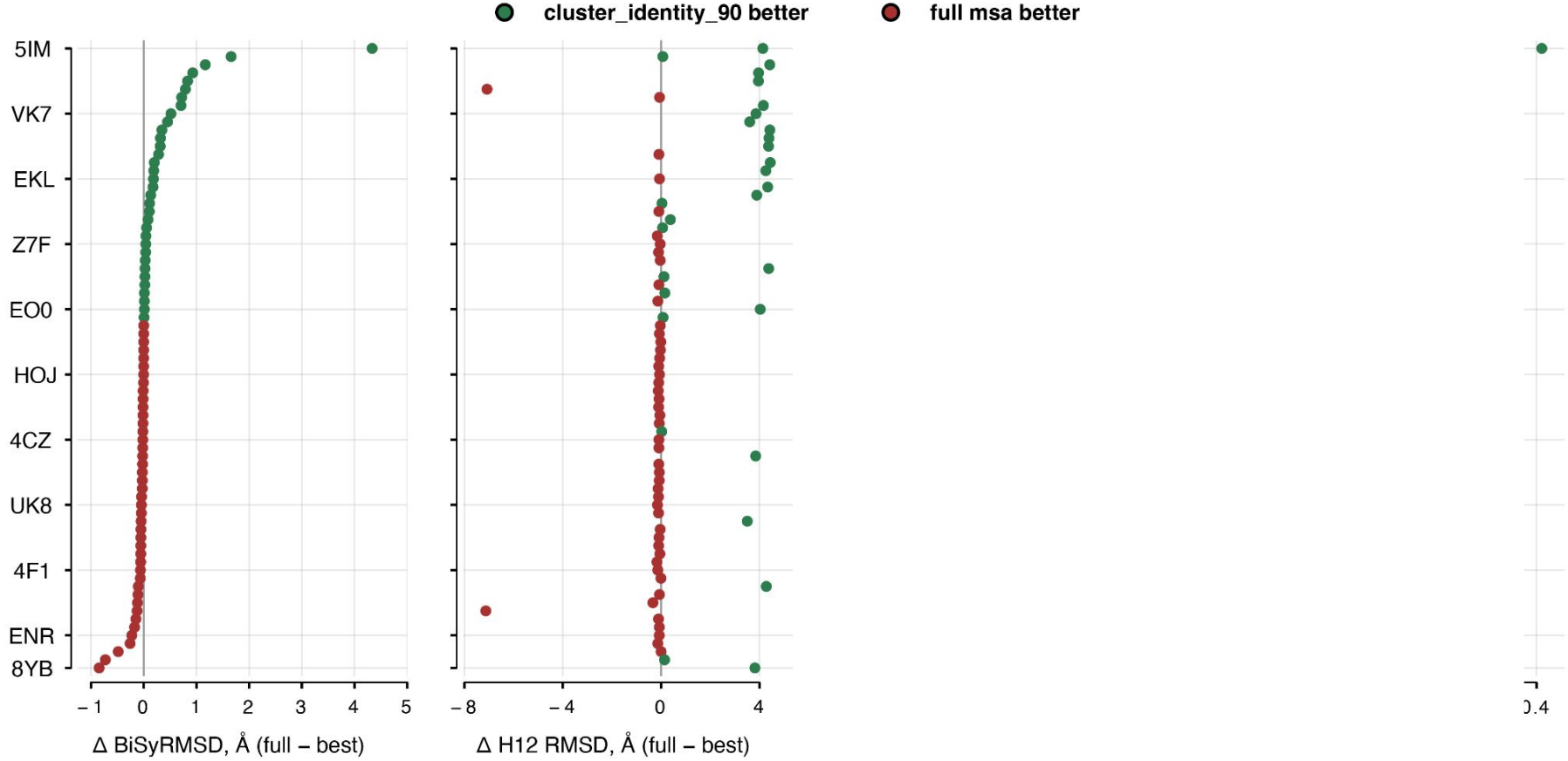
Did we find the “perfect” MSA?



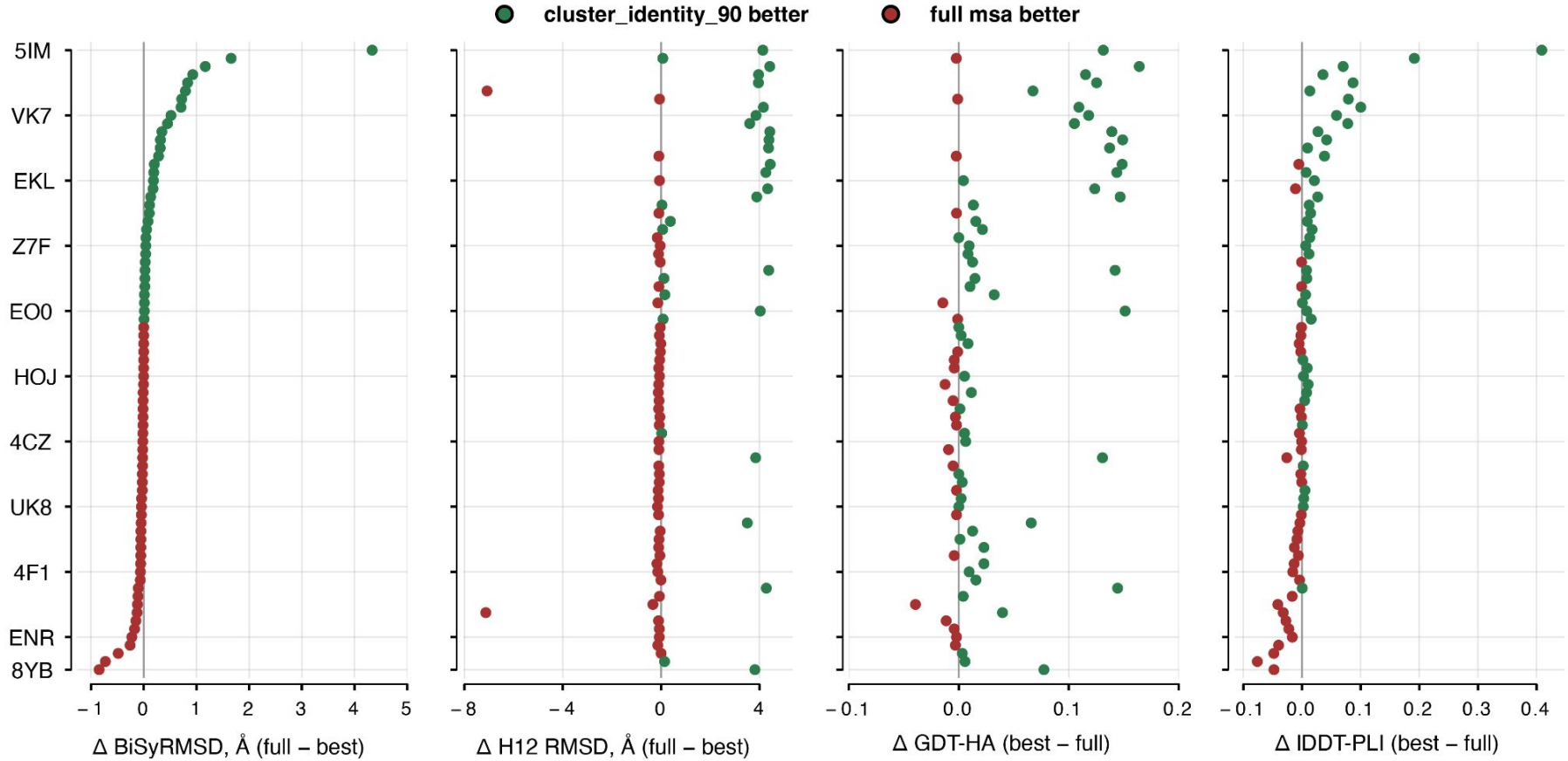
How “the perfect MSA” performs across the ligands



How “the perfect MSA” performs across the ligands



How “the perfect MSA” performs across the ligands

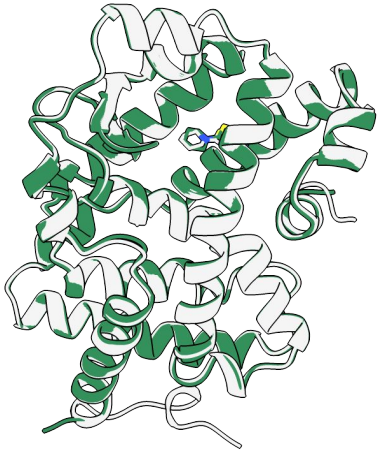


Let’s look at the structures

5G43-5IM

Crystal Structure



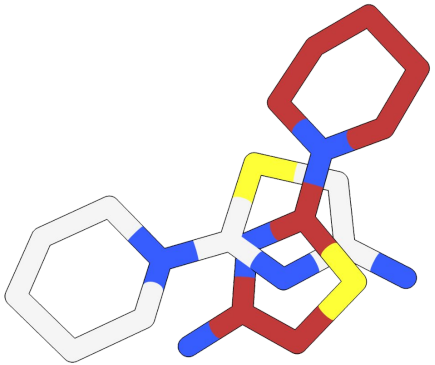


Full-MSA

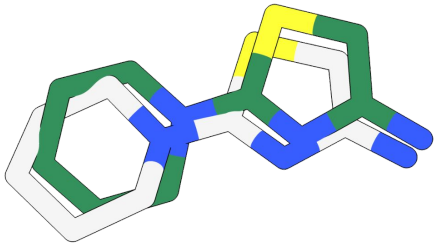
90-identity-cluster MSA

Let’s look at the structures

5G43-5IM



BiSyRMSD 5.09



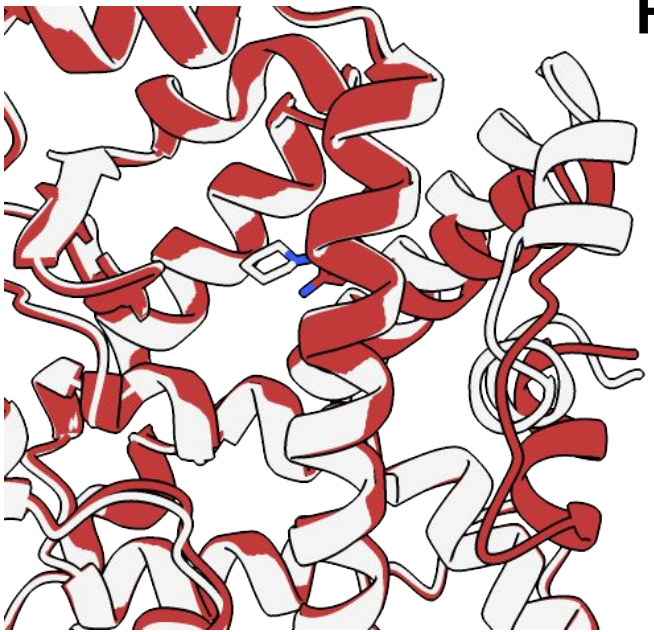
0.76

Full-MSA

90-identity-cluster MSA

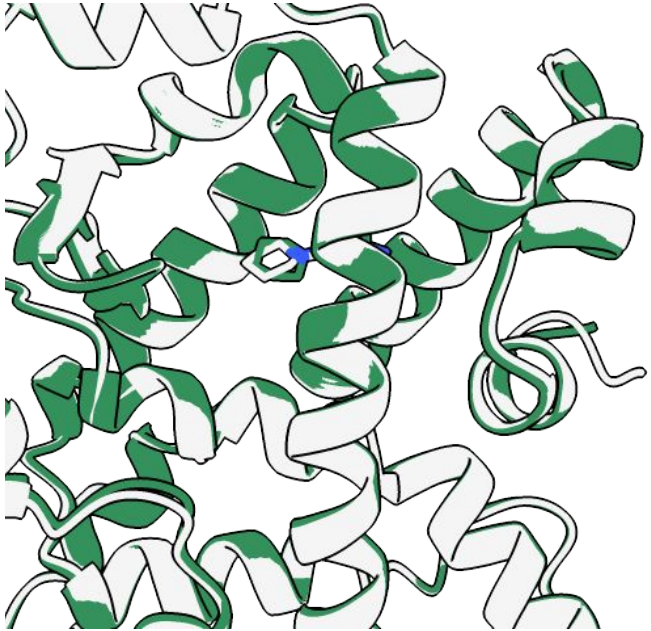
Let’s look at the structures

H12



H12 RMSD 4.73

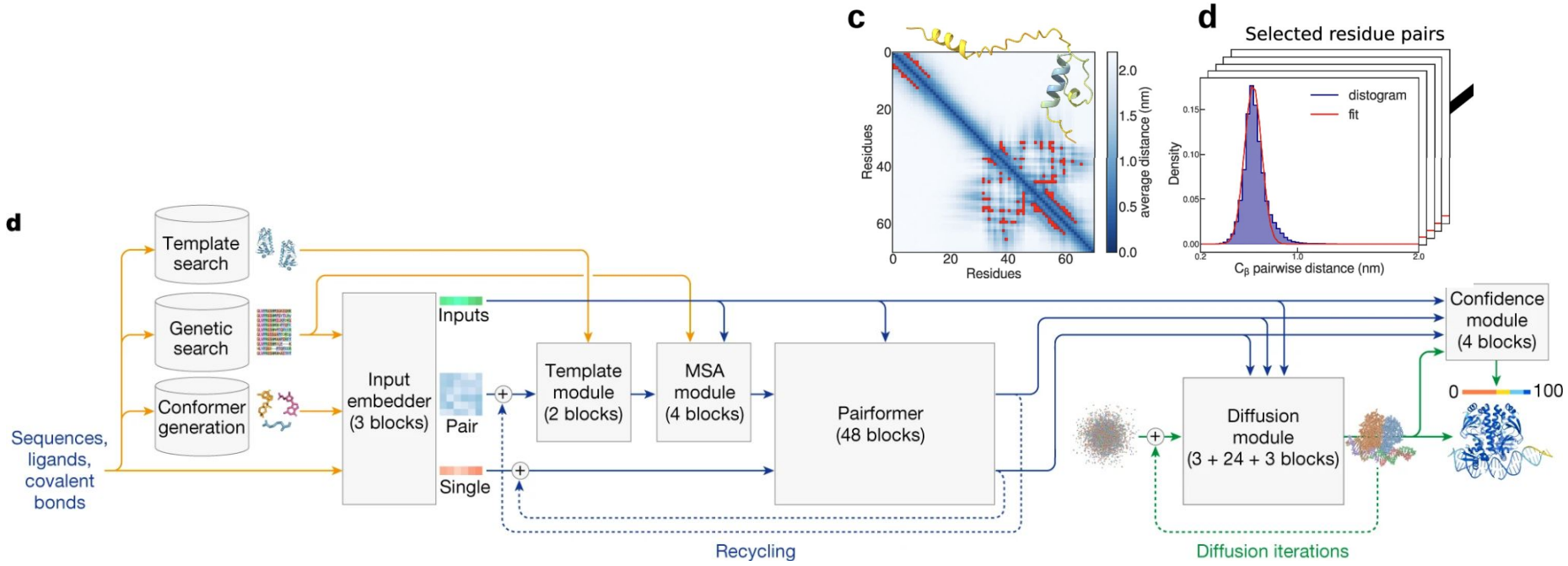
Full-MSA



0.59

90-identity-cluster MSA

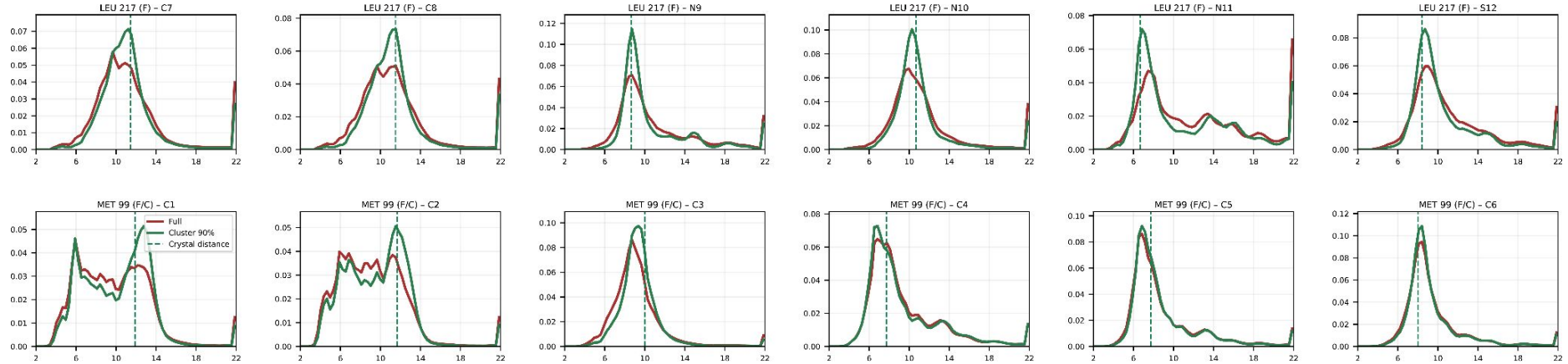
Let’s look at the distogram



Let’s look at the distogram

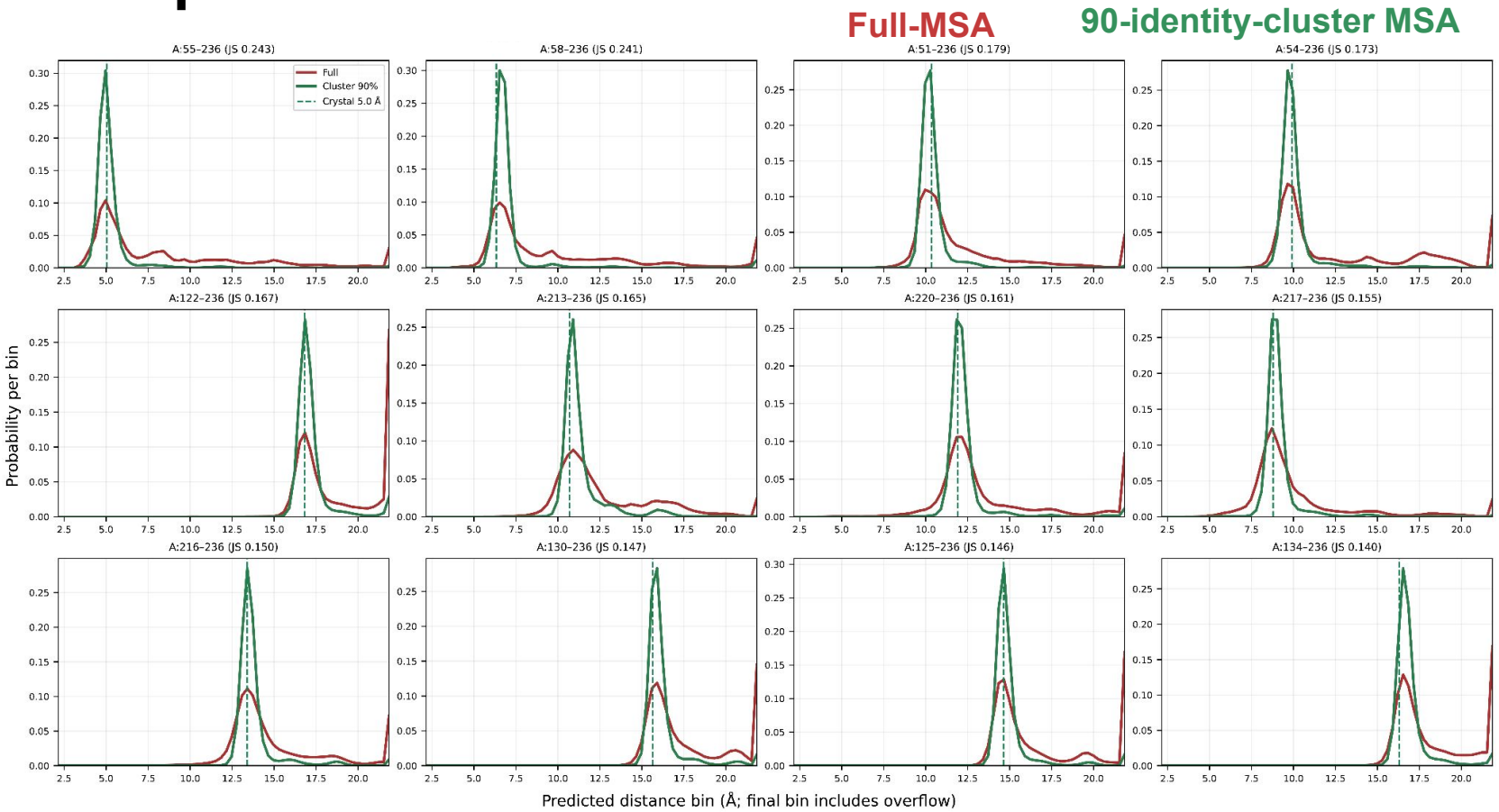
Full-MSA

90-identity-cluster MSA



5G43-5IM

“The perfect MSA” reduces co-evolution noise



Testing “the perfect MSA” on the hold-out set